

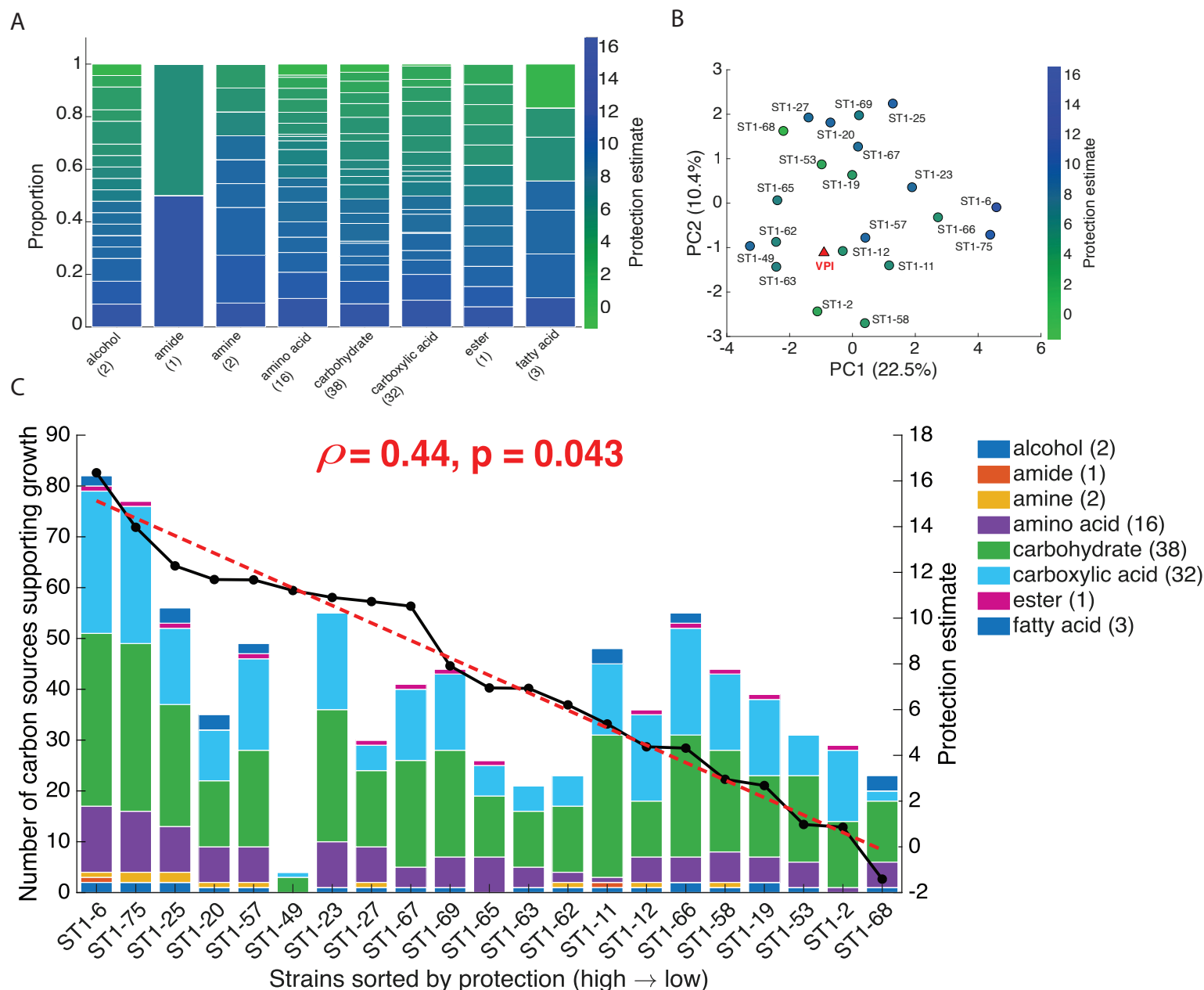


Figure 5: ST1 strains show diverse metabolic capabilities that correlate with protection. (A) Proportion of carbon sources (out of 95 BIOLOG PM1 metabolites) that supported statistically significant growth for each ST1 strain, assessed by linear mixed-effects modeling of absorbance over time relative to the negative control. Strains are colored by protection score (color scale, right). (B) Principal component analysis (PCA) of the binary growth/no-growth matrix across 95 carbon sources for all 21 ST1 strains. PC1 explains 22.5% and PC2 explains 10.4% of the variance. Strains are colored by protection score (color scale, right). (C) Number of carbon sources supporting growth for each ST1 strain, sorted from highest to lowest protection score (left to right). The black solid dots and black line indicates the protection score per strain and the red dashed line indicates the linear fit between the protection score and number of unique metabolites utilized by that strain to grow. The breadth of metabolic utilization is positively correlated with protection score.